

JEE MAIN REVISION MASTER: ELLIPSE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Eccentricity, Foci & Tangents of Ellipse	4 PYQs	Ellipse
Standard	Focal Properties & Auxiliary Circle	0 PYQs	Focal Geometry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Eccentricity, Foci & Tangents of Ellipse	$e = \sqrt{1 - \frac{b^2}{a^2}}$, $\text{Foci: } (\pm ae, 0)$, $\text{LR} = \frac{2b^2}{a}$, $\text{Tangent: } y = mx \pm \sqrt{a^2 m^2 + b^2}$, $\text{Director: } x^2 + y^2 = a^2 + b^2$
Focal Properties & Auxiliary Circle	$SP + S'P = 2a = \text{Major Axis}$, $\text{Auxiliary Circle: } x^2 + y^2 = a^2$, $\text{Parametric: } (a \cos \theta, b \sin \theta)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Eccentricity, Foci & Tangents of Ellipse (Moderate Yield — 4 PYQs)

Core Principle: Standard ellipse equation, focal distance sum property $SP + S'P = 2a$, eccentricity $e < 1$, and director circle.

■ **Caution / JEE Pitfall:** If $b > a$ (vertical ellipse), major axis is along y-axis and $e = \sqrt{1 - a^2/b^2}$ with foci at $(0, \pm be)$.

■ **Speed Trick:** Product of perpendiculars from the two foci onto any tangent to the ellipse is always constant and equal to b^2 .

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "The length of the latus rectum of $x^2 + 3y^2 = 12$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the area of the ellipse is $\frac{\pi}{225} + \frac{\pi}{2} \lambda^2 = 1$ is 20π sq units, then λ is..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "The eccentricity of the ellipse $9x^2 + 25y^2 = 225$ is..." [Ans: N/A]

3.2 Focal Properties & Auxiliary Circle (Standard Yield — 0 PYQs)

Core Principle: Focal radius property $SP + S'P = 2a$, auxiliary circle projections, and parametric coordinate representation.

■■ **Caution / JEE Pitfall:** In an ellipse, the eccentric angle θ is measured on the auxiliary circle, NOT from the center to the point on the ellipse.

■ **Speed Trick:** Maximum area of a rectangle inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $2ab$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "The length of the latus rectum of $x^2 + 3y^2 = 12$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the area of the ellipse is $\frac{x^2}{25} + \frac{y^2}{\lambda^2} = 1$ is 20π sq units, then λ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

The length of the latus rectum of $x^2 + 3y^2 = 12$ is

- (A) $(2/3)$ units
- (B) $(1/3)$ units
- (C) $\frac{4\sqrt{3}}{3}$ units
- (D) 24 units

Correct Answer: None

Solution Analysis: $\frac{x^2}{12} + \frac{y^2}{4} = 1$ L.R. $= \frac{2b^2}{a} = \frac{2 \times 4}{\sqrt{12}} = \frac{4\sqrt{3}}{3}$